# Alberto Bermejo Magarín

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## About Me

Computer engineering specialized in software engineering with a strong focus on artificial intelligence. Combines a solid software architecture foundation—with hands-on project experience in TypeScript and mobile development with React Native—alongside practical experience building end-to-end generative AI and RAG solutions during an industry internship and bachelor's thesis. Passionate about building robust, scalable, and production-ready data and AI systems.

## Experience

**Atmira** Córdoba, Spain

**Data & AI engineer intern** Feb 2026 – Jun 2026

- Developed and supported AI and data-driven initiatives aimed at delivering production-ready solutions within an agile team environment.
- Implemented data processing pipelines and generative AI features, adhering to industry best practices for scalability and software delivery.
- Completed intensive professional training programs covering advanced AI topics, effective corporate communication, and modern enterprise workflows.
- Collaborated closely with cross-functional software engineering teams to align data/AI implementations with business requirements and project goals.

## Projects

**Edge AI Airway Predictor (Hackathon IMIBIC 2025)** *Hackathon IMIBIC 2025 Post*

- Developed and deployed an edge AI system on Raspberry Pi for predicting airway type via facial recognition.
- Optimized the model pipeline, preprocessing, and inference, achieving 90% prediction accuracy and enabling anesthesiologists to anticipate intubation complications, winning the Best Device Award at Hackathon IMIBIC 2025.

**Enterprise RAG Architecture on Databricks (Bachelor's Thesis)** *Repository 1 - Repository 2*

- Built two end-to-end RAG architecture on Databricks Lakehouse, using Unity Catalog for governance and MLflow for pipeline observability.
- Implemented advanced retrieval pipelines featuring hybrid search (Dense + BM25), metadata filtering, and Cohere cross-encoder reranking.
- Evaluated system performance with RAGAS, optimizing context precision, recall, and faithfulness across multiple chunking strategies and LLMs.

**Enterprise AI Observability & Monitoring Tool (atmira)**

- Engineered an enterprise LLM observability solution to capture, trace, and monitor multi-agent workflows and AI system execution.
- Implemented automated evaluation routines and actionable optimization suggestions to improve latency, prompt costs, and response accuracy.

## Skills and technologies

**Languages & Frameworks:** Python, Java, JavaScript, TypeScript, React, React Native, SQL
**AI, Data & Cloud:** LangChain, Databricks, Firebase, Chroma
**Tools & Infrastructure:** GitHub, Docker, Kubernetes
**Languages:** Native Spanish / Cambridge C1 English / Basic French

## Education

**Universidad de Córdoba** Córdoba, Spain
BSc in Computer Engineering 2022 - 2026

**Neumann János Egyetem** Kecskemet, Hungary
Computer science exchange program. Sep 2024 – Feb 2025